

LIU HENGZHI

Email: Lihengzhi53@foxmail.com | HP: +65 89422843
LinkedIn: www.linkedin.com/in/hengzhi-liu | GitHub: github.com/53NotFifty-three

EDUCATION

National University of Singapore <i>Bachelor of Computing in Computer Science (Tuition Grant)</i>	2026 – June 2029
Temasek Polytechnic, Singapore <i>Diploma in Biomedical Engineering with Merit</i> Awarded Polytechnic Scholarship, Course Bronze Medalist, and Director's List honors.	2023 – 2026

SKILLS

Programming Languages: Python, C++, C, SQL

Systems & AI Infra: LLM Agents, System Design Basics, Embedded Systems, Real-time Systems (FreeRTOS), Firmware Development

Tools & Frameworks: Git/GitHub, Linux, FastAPI, MySQL, AWS (Aurora), CAD, Adobe Suite

TECHNICAL PROJECTS

DTx Cognitive SaaS (ADHD Therapeutics) <i>Full-Stack Developer</i> <i>React, FastAPI, AWS Aurora, LLM Agents</i> - Built a clinical ADHD SaaS platform featuring adaptive test pacing calibrated to severity. - Reduced data latency by implementing local telemetry batching with AWS token authentication. - Built an autonomous AI Copilot via Gemini API with local report caching.	2026
Smart Mahjong Play-Board <i>Independent Systems Developer</i> <i>C++, ESP32, PCB Design, Embedded Firmware</i> - Developed ESP32-based smart play-board hardware featuring custom PCB circuit integration. - Achieved lag-free, real-time physical-to-digital state updates by writing optimized ESP32 C++ firmware.	2026

WORK EXPERIENCE

National University of Singapore (iHealthtech) <i>Research Assistant Intern</i> <i>Singapore</i> - Generated high-quality experimental datasets for biomedical research by setting up microscale testing workflows. - Ensured data integrity across all testing pipelines by performing comprehensive validation checks on hardware instruments. - Improved data traceability by establishing structured digital documentation standards for all experimental records.	Mar 2025 – Jul 2025
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CO-CURRICULAR ACTIVITIES

WorldSkills Singapore (Additive Manufacturing) Won 1st Place (TP Selection Round) by modeling and optimizing 3D printable products.	2024
Electronic Design & Robotics Competitions Developed embedded logic systems and real-time control algorithms for robotics challenges.	2024